SEOUL BIKE RENTAL DEMAND ANALYSIS

Data Analysis & Visualization Portfolio Project

An exploratory analysis of hourly rental patterns, seasonal variation, and weekday-versus-weekend demand using Google Sheets and Tableau Public.

TOTAL RENTALS	AVG. RENTALS / OPERATING HOUR	PEAK HOUR
6,172,314	**729.16**	**6:00 PM**
Dataset total shown in dashboard	Average shown in dashboard	1,554.02 average rentals

Project at a glance

- **Objective:** identify peak and low-demand hours, compare seasonal patterns, and examine weekday/weekend differences.
- **Tools:** Google Sheets for data preparation, formulas, and Pivot Tables; Tableau Public for dashboard visualization.
- **Deliverables:** analysis workbook, interactive dashboard, and this project report.

This report describes observed patterns in the supplied Pivot Table results. Business recommendations are preliminary and should be validated with additional operational data.

Executive Summary

This project explores Seoul bicycle rental demand, focusing on time-of-day patterns, seasonal variation, and differences between weekdays and weekends. Google Sheets was used to organize and summarize the dataset, while Tableau Public was used to present the results in a dashboard.

Finding	Evidence	Interpretation
Evening peak	6:00 PM: 1,554.02 average rentals; 8:00 AM: 1,050.23.	Demand is concentrated around specific hours; commuting is a possible explanation but is not confirmed by rental counts alone.
Seasonal variation	Summer: 1,034.07; Autumn: 924.11; Spring: 746.25; Winter: 225.54.	Summer has the highest seasonal average and Winter the lowest.
Weekday/weekend	Weekday 8:00 AM: 1,306.02 vs weekend 420.79; weekend 3:00 PM: 1,022.60 vs weekday 790.23.	Demand patterns vary by hour and day type.

The findings can inform further evaluation of bicycle availability, seasonal fleet readiness, and different operating plans for weekdays and weekends. The analysis does not identify station-level shortages or prove causal relationships.

1. Dataset & Data Preparation

The supplied Seoul bicycle rental dataset contains 8,760 hourly observations and 14 original columns. Fields include rental counts, date and hour information, seasons, holiday status, functioning-day status, and weather-related measurements.

Preparation workflow

- Imported the dataset into Google Sheets and created a separate working sheet for analysis.
- Derived Day of Week from the date field.
- Classified each observation as Weekday or Weekend.
- Created Hour Category using a lookup table and VLOOKUP with approximate matching.
- Created Operational Status from the Functioning Day indicator.
- Built Pivot Tables for hourly averages, hourly averages by season, and hourly averages by day type.

Data handling note: this report documents the preparation steps described for this project. It does not claim that missing values, duplicates, or outliers were removed unless those checks were explicitly performed.

2. Analytical Approach

Google Sheets

Google Sheets was used for calculated columns, lookup logic, Pivot Tables, and KPI summaries. The main measures examined were total rentals and average rental count by hour, season, and day type.

Tableau Public

The dashboard combines three KPI summaries with an hourly demand line chart, a day-of-week-by-hour heatmap, and a seasonal comparison chart. These views provide complementary perspectives on time-based and seasonal demand.

Metric interpretation

The seasonal and day-type comparisons use average rental count values shown in the Pivot Tables. The total rentals KPI is a sum. These metrics describe different aspects of demand and should not be compared as if they were the same measure.

3. Key Findings

3.1 Evening and Morning Peaks

Average rental demand reached 1,554.02 at 6:00 PM, the highest value in the hourly Pivot Table. A secondary peak occurred at 8:00 AM, with 1,050.23 average rentals. The lowest hourly average occurred at 4:00 AM, at 137.49.

Interpretation: rental activity varies considerably across the day. The morning and evening peaks may be consistent with commuting patterns, but travel purpose cannot be confirmed from rental counts alone.

3.2 Seasonal Variation

Season	Average rentals
Summer	1,034.07
Autumn	924.11
Spring	746.25
Winter	225.54

Summer had the highest overall average and Winter the lowest. The Summer average was approximately 4.58 times the Winter average, indicating substantial seasonal variation in observed rental counts.

3.3 Weekday vs Weekend

Hour	Weekday avg.	Weekend avg.	Higher value
8:00 AM	1,306.02	420.79	Weekday
12:00 PM	700.09	780.13	Weekend
3:00 PM	790.23	1,022.60	Weekend
6:00 PM	1,724.05	1,135.60	Weekday

Weekday averages were higher during the morning and evening peaks, while weekend averages exceeded weekday averages at several midday hours. Overall averages were 748.11 for weekdays and 682.38 for weekends.

3.4 Seasonal Variation at the Evening Peak

At 6:00 PM, average rentals were 2,135.14 in Summer, 1,983.33 in Autumn, 1,689.31 in Spring, and 438.30 in Winter. The evening peak was present in all four seasons, but its magnitude differed substantially.

4. Business Recommendations

Review peak-hour availability

Evaluate bicycle availability and redistribution requirements before the morning and evening peaks. Use station-level data to identify where additional capacity may be needed.

Incorporate seasonality

Use seasonal demand patterns to inform fleet readiness and maintenance planning. Consider testing targeted promotions during low-demand periods and measure their effectiveness.

Differentiate weekday and weekend plans

Review whether redistribution schedules should differ between weekdays and weekends, particularly around morning, evening, and midday periods.

Evaluate low-demand windows

Assess whether selected maintenance or redistribution activities can take place during lower-demand hours without reducing service availability.

Validate before implementation

Combine hourly results with station-level demand, bicycle availability, maintenance records, and operational costs before implementing changes.

5. Limitations

- The analysis describes observed rental patterns and does not establish causal relationships.
- Aggregate hourly averages do not identify specific stations experiencing shortages or excess inventory.
- Weather variables are present in the dataset, but their individual effects were not tested in the reported analysis.
- Operational recommendations are preliminary and should be validated with additional data before implementation.

6. Conclusion

This project used Google Sheets and Tableau Public to explore Seoul bicycle rental demand by hour, season, and day type. The analysis identified an evening peak at 6:00 PM, substantial seasonal variation, and distinct weekday/weekend patterns. The findings demonstrate how spreadsheet-based preparation, Pivot Tables, KPI summaries, and visual dashboards can support exploratory analysis and communicate operational patterns.

Further analysis using station-level information and additional operational variables would help determine where specific service changes could have the greatest impact.

Project Deliverables

- Google Sheets workbook containing data preparation, formulas, Pivot Tables, and KPI summaries.
- Tableau Public dashboard for exploring hourly, seasonal, and day-type demand patterns.
- This report documenting the analytical approach, key findings, recommendations, and limitations.